Luiss
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degli Studi Sociali Guido Carli

Homework Exercises on Binary Numbers and Boolean Logic

Introduction to Computer Programming

Management and Artificial Intelligence

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Binary Number Exercises

Decimal 387 to Binary

Convert the decimal number **387** to binary:

$387 \div 2 = 193$ (rem 1)

$193 \div 2 = 96$ (rem 1)

$96 \div 2 = 48$ (rem 0)

$48 \div 2 = 24$ (rem 0)

$24 \div 2 = 12$ (rem 0)

$12 \div 2 = 6$ (rem 0)

$6 \div 2 = 3$ (rem 0)

$3 \div 2 = 1$ (rem 1)

$1 \div 2 = 0$ (rem 1)

Result: $(110000011)_2$

Binary 110110101011 to Decimal

Convert the binary number **110110101011** to decimal:

$$
\begin{aligned}
(110110101011)_2 &= 1 \cdot 2^{11} + 1 \cdot 2^{10} + 0 \cdot 2^{9} + 1 \cdot 2^{8} + 1 \cdot 2^{7} + 0 \cdot 2^{6} + 1 \cdot 2^{5} + 0 \cdot 2^{4} + 1 \cdot 2^{3} \\
&= 2048 + 1024 + 0 + 256 + 128 + 0 + 32 + 0 + 8 + 0 + 2 + 1 = (3499)_{10}
\end{aligned}
$$

Decimal 2048 to Hexadecimal

Convert the decimal number **2048** to hexadecimal:

$2048 \div 16 = 128$ remainder 0

$128 \div 16 = 8$ remainder 0

$8 \div 16 = 0$ remainder 8

Reading the remainders from bottom to top: 800

Therefore, $2048_{10} = 800_{16}$

Hex BADC0DE to Binary

Convert the hexadecimal number **BADC0DE** to binary:

Convert each hexadecimal digit to its 4-bit binary equivalent:

- $B_{16} = 1011_2$
- $A_{16} = 1010_2$
- $D_{16} = 1101_2$
- $C_{16} = 1100_2$
- $0_{16} = 0000_2$
- $D_{16} = 1101_2$
- $E_{16} = 1110_2$

Therefore: $BADC0DE_{16} = 1011101011011100000011011110_2$

Text TOTTI to Binary

Convert the text "TOTTI" to binary using the ASCII table

```
- T = 84 in ASCII = 01010100 in binary
- O = 79 in ASCII = 01001111 in binary
- T = 84 in ASCII = 01010100 in binary
- T = 84 in ASCII = 01010100 in binary
- I = 73 in ASCII = 01001001 in binary
```

Binary to text

Convert `0100110001010101010010010101001101010011` into text using ASCII:

Breaking the binary string into 8-bit chunks:

- 01001100 -> 76 in decimal -> 'L' in ASCII
- 01010101 -> 85 in decimal -> 'U' in ASCII
- 01001001 -> 73 in decimal -> 'I' in ASCII
- 01010011 -> 83 in decimal -> 'S' in ASCII
- 01010011 -> 83 in decimal -> 'S' in ASCII

So the text is: LUISS

Boolean Logic Exercises

Formula $(A \land \neg B \land C) \lor (\neg A \land B \land \neg C) \lor (A \land B \land C)$

A	B	C	!A	!B	!C	A∧!B∧C	!A∧B∧!C	A∧B∧C	(A∧!B∧C)∨(!A∧B∧!C)∨(A∧B∧C)
0	0	0	1	1	1	0	0	0	0
0	0	1	1	1	0	0	0	0	0
0	1	0	1	0	1	0	1	0	1
0	1	1	1	0	0	0	0	0	0
1	0	0	0	1	1	0	0	0	0
1	0	1	0	1	0	1	0	0	1
1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	0	0	1	1

Formula $(\neg A \wedge \neg B \wedge \neg C) \vee (A \wedge B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

A	B	C	!A	!B	!C	!A∧!B∧!C	A∧B∧C	A∧!B∧!C	(!A∧!B∧!C)∨(A∧B∧C)∨(A∧!B∧!C
0	0	0	1	1	1	1	0	0	1
0	0	1	1	1	0	0	0	0	0
0	1	0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0	0	0
1	0	0	0	1	1	0	0	1	1
1	0	1	0	1	0	0	0	0	0
1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	0	1	0	1

Formula $(A \wedge B) \vee (\neg A \wedge \neg B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

A	B	C	!A	!B	!C	A∧B	!A∧!B∧C	A∧!B∧!C	(A∧B)∨(!A∧!B∧C)∨(A∧!B∧!C)
0	0	0	1	1	1	0	0	0	0
0	0	1	1	1	0	0	1	0	1
0	1	0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0	0	0
1	0	0	0	1	1	0	0	1	1
1	0	1	0	1	0	0	0	0	0
1	1	0	0	0	1	1	0	0	1
1	1	1	0	0	0	1	0	0	1

Formula $(A \wedge B \wedge C) \vee (\neg A \wedge \neg B \wedge \neg C)$ is equivalent to what?

A	B	C	A∧B∧C	!A	!B	!C	!A∧!B∧!C	(A∧B∧C)∨(!A∧!B∧!C)
0	0	0	0	1	1	1	1	1
0	0	1	0	1	1	0	0	0
0	1	0	0	1	0	1	0	0
0	1	1	0	1	0	0	0	0
1	0	0	0	0	1	1	0	0
1	0	1	0	0	1	0	0	0
1	1	0	0	0	0	1	0	0
1	1	1	1	0	0	0	0	1

Formula $(A \vee B) \wedge \neg(A \wedge B)$ is a tautology?

A	B	A∨B	A∧B	!(A∧B)	(A∨B)∧!(A∧B)
0	0	0	0	1	0
0	1	1	0	1	1
1	0	1	0	1	1
1	1	1	1	0	0

No, it is not a tautology! This is actually the XOR operation.

Formula $\neg A \wedge \neg B \wedge A$ is a contradiction?

A	B	!A	!B	!A∧!B∧A
0	0	1	1	0
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

Yes, it is a contradiction! It requires A to be both true and false simultaneously.